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Article

The Study of Metal−Carbonyl Complexes by Means of Computational IR Spectra Analysis: A Remote Didactic Approach Based on Chemical Thinking

Miguel Reina, Itzel Guerrero-Ríos, and Antonio Reina*

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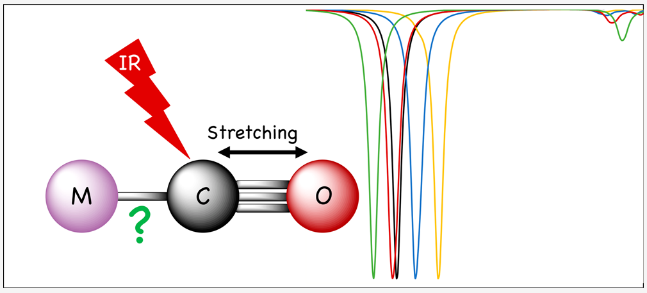
ABSTRACT: We describe a remote pedagogical approach based on chemical thinking to study metal−carbonyl complexes by analyzing simulated IR spectra. The proposed approach, implemented due to the COVID-19 pandemic, can be employed in classrooms that have very limited laboratory equipment for evaluating toxic metal−carbonyl compounds, as well as for synthesizing compounds that have not been reported. The method, consisting of a class lecture accompanied by a remote computational activity, aims to provide students with the ability to assemble concepts from different fields, such as organometallic chemistry and analytical chemistry, while taking advantage of computational methods to answer higher level questions. We evaluated whether analyzing the nature of M−CO bonding was appropriate for achieving these educational goals. Octahedral compounds of the $M(CO)_6$ and $M(CO)_4L_2$ type, bearing a variety of metal centers (M = Cr, Mo, W, V, Mn and Fe) and ligands (L = phosphines and phosphites), as well as bimetallic $Fe_2(CO)_9$, were compared, showing how these modifications affect M−CO bonding. After the didactic session, attended by second-year and upper-division students of Facultad de Química at UNAM, an evaluation and survey showed that students improved their understanding of the subject when they obtained and visualized IR spectra, also exhibiting greater confidence and enthusiasm for addressing challenging topics. The combination of computational results, spectroscopic analysis, and organometallic theory represents an efficient and clear procedure for implementing chemical thinking, regardless of the difficulties posed by the COVID-19 pandemic.



KEYWORDS: *Second-Year Undergraduate, Upper-Division Undergraduate, Inorganic Chemistry, Organometallics, IR Spectroscopy, Computational Chemistry*

■ INTRODUCTION

Government strategy to control COVID-19 pandemic required the population to preserve mutual healthy distance and stay at home.[1] Lockdown had evident negative effects on education, in particular the teaching of chemistry, as laboratories constitute an important source of empirical knowledge that helps theoretical learning to be more easily integrated. In addition, in traditional chemistry teaching, academic subjects are sometimes presented as isolated or highly compartmented topics.[2,3] Yet in reality, these topics are interconnected, and when combined, more intricate scientific issues become clear.[4] In general, students study related material as separate subjects. Therefore, following the traditional approach to learning chemistry, students strive to mobilize the different acquired concepts, as these are categorized as apparently separate topics. In this context, students are unable to either answer challenging questions or solve queries beyond the most basic academic content, as is particularly apparent, among higher level undergraduates. In particular, upper-division students from the Facultad de Química, at Universidad Nacional Autónoma de México (UNAM) in Mexico struggle to explain intricate concepts and solve complex problems. Advanced inorganic chemistry courses (i.e., Organometallic Chemistry, Bioinorganic Chemistry, Homogeneous Catalysis) provide an example; here students are required to associate concepts from subjects they studied previously, involving different fields of chemistry. In this regard, organometallic chemistry lessons are challenging for students because chemical bonding theories, analytical chemistry, organic reactions, periodic trends, and atomic structure concepts are necessary if the students are to thoroughly understand both the bonding and reactivity of metal−ligand bonds. In particular during the last years, some reports have focused on the implementation of organometallic chemistry courses for undergraduates.[5−9] Authors pointed out

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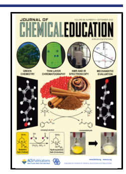

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the difficulty to fully comprehend organometallic topics since students need to master several important concepts.

In the Organometallic Chemistry arena, transition metal compounds present wide and important applications in electronics, agrochemicals, pharmaceuticals, and the fuel industry, among others;[10−14] consequently, their study is of particular interest in any undergraduate chemistry program. One of the most common ligands in transition metal organometallic complexes is the carbonyl ligand, CO.[15−26] There are many neutral, anionic, and cationic binary metal carbonyls, for which carbonyl is the only ligand around the metal center, and there exists an even larger number of compounds of metal carbonyls containing organic and inorganic ligands.[19−26] In this sense, transition metal carbonyl compounds are exceptional representatives of organometallic chemistry, and in addition, infrared (IR) spectroscopy, a rather inexpensive and readily available technique for characterization of transition metal carbonyls,[17,27−29] is an ideal teaching tool to introduce students to the fundamental electronic structure aspects involved in M−CO bonding (M = transition metal). However, its use depends on having a laboratory with an IR instrument and having the pure substance on hand. So, if one is interested in comparing the stretching carbonyl region of the IR spectra (ν(CO)) of a large set of carbonyl complexes with different metals and/or different ligands, their preparation and purification could be an expensive and time-consuming task. Plus, some metal carbonyls such as $Ni(CO)_4$, $Fe(CO)_5$, and $Cr(CO)_6$ are volatile, highly toxic, and must be carefully handled.[30−32] In overall and to properly address this intricate topic, a high level of abstraction is required to get a good perception of the nature of the M−CO bonding: molecular orbital analysis and knowledge of infrared spectroscopy are handy to establish the contribution between donations and back-bonding between the metal and the CO group.

At this point, the Dewar-Chatt-Duncanson (DCD) model can be very useful to separately study the components of metal−ligand bonding, σ-donation, and π-backbonding.[33,34] The use of this model may allow educators to introduce and study several aspects including symmetry analysis, group theory to predict the number and nature of infrared active CO stretching vibrations, and quantification of σ and π components in metal−ligand bonding, among others.[35−39] While these topics are of huge interest and may enrich the discussion about M−CO bonding, important background knowledge, such as notions of symmetry, is necessary. Unfortunately, in our faculty's curricula, symmetry is barely treated among undergraduate courses. For this reason, we designed an activity to understand the M−CO bond in which symmetry and group theory are not discussed. However, we strongly believe that the proposed activity will be helpful to introduce highly challenging topics, in particular because the students learn to mobilize different concepts by developing chemical thinking. Moreover, we encourage educators that may use our didactic proposal, to take advantage of the DCD model in order to do so.

In this study, we present a pedagogical approach in the form of a class lecture and a computational remote activity employed during the COVID-19 pandemic. Noteworthy, this remote teaching strategy enables the study of metal−carbonyl complexes overcoming the impossibility to attend the teaching laboratory to synthesize, purify, and analyze the samples. Moreover, the computational approach allows the students to analyze many metal carbonyls and to visualize the ν(CO) region of the IR spectra, contrary to the usual tabulated maximum wavelength values. Octahedral $M(CO)_6$ and trans-$M(CO)_4L_2$ compounds (M = Cr, Mo, W, V, Mn, and Fe) and ligands (L = phosphines and phosphites) were studied. This approach is focused on chemical thinking,[2] and it aims to furnish students with versatile and diverse learning tools (chemical bonding theory, IR spectra analysis, and computational chemistry advantages), all combined to address the nature of the M−CO bond. Likewise, this didactic proposal can be transposed to contexts where organometallic chemistry is attempting to gain relevance, but where the appropriate laboratory equipment is not available. In this regard, computational chemistry has reached a point wherein IR spectra simulation of a plethora of carbonyl complexes can be achieved under safe conditions without using any chemical reagent. It has been established that easily accessible calculations and predictions of spectroscopic data of carbonyl compounds is a powerful tool in teaching group theory,[40] evaluating Fischer and Schrock carbenes,[41] and studying effects concerning the metal nature in carbonyl complexes.[42]

Finally, and as a proof of concept, the lecture and the remote computational activity were held for two different groups of students. The first group comprised students, in their last undergrad year, enrolled in the 3-h weekly Homogeneous Catalysis elective course, and the second group consisted of volunteer students enrolled in the mandatory second-year 3-h weekly Inorganic Chemistry subject. The main objective of this was to demonstrate that regardless of the level of the students, the didactic proposal and the use of a computational strategy could be useful in learning the fundamental electronic aspects involved in M−CO bonding. All participating students belong to the Facultad de Química of UNAM and were evaluated prior to and after the lecture and the remote activity. Resulting data indicated that this remote didactic approach might improve the understanding of metal−carbonyl bonding during COVID-19 pandemic; however more data are required to reach a definitive conclusion. In spite of this, the student survey applied after the lecture revealed that studying metal−carbonyl bonding with reference to IR spectra was helpful because it gave students the confidence to predict and understand the electronic effects that affect metal−carbonyl bonding.

■ DIDACTIC APPROACH

Lecture

The didactic proposal consisted of a particular format lecture designed to hold students' attention and promote discussion, and a remote computational activity in which the students were able to visualize the optimized geometries of the different compounds that were studied, analyze the IR spectra obtained computationally, and examine the different vibration modes. The main goal of both the lecture and the activity was to answer the following questions assertively.

- What is the nature of the CO ligand in terms of σ/π acceptance/donation?
- What is the actual meaning of the vibration frequency associated with CO stretching band in the IR spectra of a metal carbonyl compound?
- What is the effect, on the CO stretching IR band, of having a ligand different to CO in a trans-$M(CO)_4L_2$-type complex?

- Which kind of ligands will strengthen the M−CO bond and which ones will strengthen the C−O bond?
- Is it possible to predict how the carbonyl stretching vibration will be modified depending on the metal or ligand nature?

In what follows, the characteristics of the lecture and the remote computational activity will be explained. In Table 1, all

Table 1. Studied Systems

Octahedral $M(CO)_6$		
Neutral	Cationic	Anionic
$Cr(CO)_6$; $Mo(CO)_6$ and $W(CO)_6$.	$[Fe(CO)_6]^{2+}$ and $[Mn(CO)_6]^+$	$[V(CO)_6]^-$
Octahedral *trans*-$M(CO)_4L_2$		
Phosphines		Phosphites
phosphine, triphenylphosphine, triethylphosphine, trichlorophosphine, tris(4-nitrophenyl)phosphine, tris(4-methoxyphenyl)phosphine		trimethylphosphite and triphenylphosphite

the studied systems are presented. The lecture begins with an introduction about carbonyl ligand and M−CO bonding by interpretation of the molecular orbital diagram, and the possible expected changes in IR spectroscopy due to donation and backbonding is discussed. Next, periodic trends concerning binary metal carbonyls are studied through their changes in IR spectra. In this regard, the spectra analysis requires mobilization of several atomic properties such as radii, orbitals diffusion, and effective nuclear charge with the understanding of periodic anomalies observed in transition metals. Then, the lecture continues with the examination of the electronic effects of phosphine and phosphite ligands. Finally, the prediction of the behavior of a bimetallic carbonyl compound, $Fe_2(CO)_9$, allows a comparison of the M−CO bonding differences between bridging and terminal carbonyls. The main feature of the lecture format was that students had to fill in the blanks with keywords to understand the different phenomena. Some examples are given below, but the reader can find the entire lesson in the Supporting Information file (The lesson has been translated from Spanish and here is presented only in English).

Example 1: Regarding Carbonyl Ligand and M−CO Bonding. Organometallic compounds are defined as systems with at least one direct bond between a metal and a _____ atom. Those M−C bonds can be polarized depending on both metal and carbonated ligand nature. In the context of introductory organometallic chemistry class, the molecular orbital diagram of CO is an adequate starting point because it provides useful information to understand metal−carbonyl bonding (see Figure 1). It is important to notice that the $\sigma^*_{s/p}$ orbital results from the interaction of _____ atomic orbitals __ and __ and _______ atomic orbitals __ due to the low energy difference between C_{2s} and O_{2p} (3.5 eV). This interaction, called ________, destabilizes the _ molecular orbital (____), shifting it to a higher energy above the __ orbitals (____), resulting in a σ−π crossover.

Example 2: Concerning the Periodic Trends: Metal Modification. In this case there are two remarks to consider: (i) the similar ______ ______ of Mo and W, as a result of the lanthanide contraction affecting the sixth period; and, (ii) an ________ on the principal quantum number _ of the d orbitals involved in ___________ to CO and as a result, more ______ orbitals that _______ the overlap with the ______ of CO.

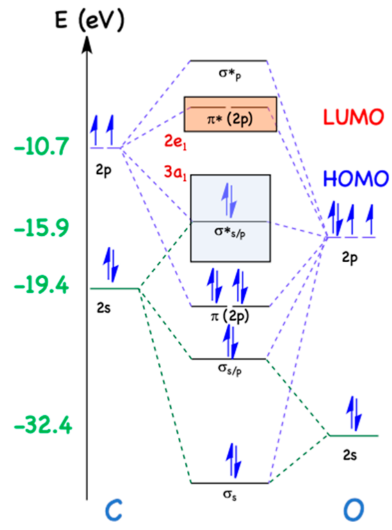

Figure 1. Molecular orbital diagram of CO.

Example 3: Examining the Effect of Different Ligands. The electronic nature of the metal strongly affects the C−O bond order of carbonyl ligands in M−CO complexes and is modulated by the ______ bonded to the metal. Phosphorus-based ligands favor __-donations and __-acceptance in metal complexes depending on the ______ of phosphorus. Different trans-$Cr(CO)_4(L)_2$octahedral complexes were evaluated, in which L corresponds to phosphorus-based ligand and occupies the apical position.

Example 4: Concluding the Lecture. Some trends about carbonyl stretching frequency (ν(CO)) in metal−carbonyl complexes can be drawn regarding the electronic nature of the complex:

- Within a group, ν(CO) ________ as the principal quantum number _______ resulting from a _____ back-donation due to the combination of lanthanide contraction and more effective overlap between metal d-orbital and CO π^* orbital.
- ν(CO) ________ within a period for ___________ complexes, as the effective nuclear charge ________.
- In general, ν(CO) ________ in metal complexes with strong _-_______ ligands and __________ for complexes with strong _-_________ ligands.
- ν(CO) is higher for _______ carbonyls than ________ carbonyls resulting from the bonding between one carbonyl ligand and two metal centers.

Although the format of the lesson is very simple, this led to interesting group discussions about what words should be placed in the blanks and, above all, it led the students to question, discuss, and understand the fundamental electronic phenomenon of systems. Overall, the lesson format was very helpful in keeping students motivated, interested, and focused during the online classes as the student survey in the next sections will show.

Remote Computational Activity

In the remote computational activity, the students learned from the optimized geometries to generate the IR spectra and to visualize the most important vibrational modes for each system. The Gaussian 09 package of programs was used for all the electronic structure simulations.[43] M06 functional[44] was employed, in conjunction with the 6-31G(d) basis set for all

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the nonmetal atoms (C, O, H, P, N, and Cl), and for metal atoms, the SDD pseudopotential was used. In all cases, harmonic analyses were calculated to verify local minima (absence of imaginary frequencies). In this context, computational calculations and their results take on greater importance, as first they solve the impossibility of obtaining transition metal carbonyl IR spectra experimentally, and second, they help in visualizing the vibration modes and identifying the predominant contribution of the different CO ligands for the IR active bands. It is important to outline, that students have not performed any of the presented calculations, since, in the context of the student's curricula at UNAM, theoretical and computational chemistry is an optional course. Theoretical calculations allow both to circumvent the COVID-19 pandemic and to study many metal carbonyls, some of them known for their toxicity, and some of them yet to be synthesized. Also, we emphasize that although many students assume that computational results are infallible, some important limitations arise from the choice of DFT package,[45] density functional[46] basis set, or pseudopotential,[45,47] and these together with convergence criteria can all have a significant effect on any calculated spectra.

In Figure 2, the IR spectra within a group comparing $Cr(CO)_6$, $Mo(CO)_6$, and $W(CO)_6$ are presented. The main explanation of this spectra can be found in the lecture presented in the Supporting Information. All other figures corresponding to the computationally obtained IR spectra are presented in Figures S4 to S9. Also videos have been included showing the most important vibrational modes of each system.

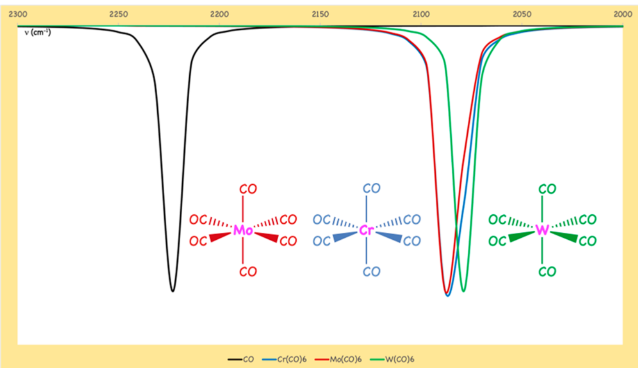

Figure 2. Calculated IR spectra of $Cr(CO)_6$, $Mo(CO)_6$, and $W(CO)_6$. (Level of theory: M06 functional and 6-31G(d) for basis set for C and O atoms, and for Mo, Cr and W atoms, SDD pseudopotential was used).

STUDENT'S FEEDBACK

To assess this remote didactic approach, the present material was used for the Homogeneous Catalysis elective course (12 students), and Inorganic Chemistry subject (30 volunteer students). A brief diagnostic assessment was performed to evaluate student's knowledge before the seminar (Figure 3 and Table S3). As has been mentioned in this work and has been continuously reported, organometallic chemistry is a discipline that requires integration of knowledge, so even for the students who took the course, the results are quite poor (50% correct answers for both). The results of the volunteer students of Inorganic Chemistry are as expected, because they barely have notions to make correct decisions. In that sense, their results serve as a retrospective testimony of the usefulness of the activities

After the lecture, Homogeneous Catalysis students slightly improved their understanding about the subject, as reflected by the postseminar evaluation (Table 2). Contrastingly, Inorganic Chemistry students show an outstanding improvement. This can be explained because the results of the first survey were very insufficient and the knowledge learned was not refuted, but rather consolidated. In that sense, these students did not have to deal with previous alternative ideas. For all questions the results are better for sophomore volunteers than for seniors. This inevitably makes us question what happens to students as the semesters and courses go by. Moreover, questions 1−4, in the afterward evaluation, showed an increment of correct answers compared to their previous IR spectra prediction of metal carbonyl complexes for all the students. After-lecture discussion and remote computational activity proved to be very important since students were able to explain in their own words why a metal carbonyl complex has a higher or lower CO stretching frequency depending on the nature of the metal and the ligands bonded.

More importantly, Figure 4 shows the results of an exit survey, which points out that students conceptualized better how metal backbonding is affected, by analyzing the infrared spectra. Moreover, they felt more confident facing a challenging topic. The survey consisted of four questions in a Likert scale. (Tables S4−S6 in the Supporting Information show the complete results). Survey analysis showed that students agreed with the first statement and consider metal−carbonyl bonding a challenging topic. In fact, undergraduates expressed, after lecture discussion, that many electronic aspects should be considered to predict and interpret IR spectra of metal carbonyl complexes and that the lecture helped them to rationalize those aspects. Moreover, most of students considered that the interpretation of infrared spectra facilitates the task, and finally, the majority of the respondents felt more confident to predict and to interpret electronic effects on metal carbonyl bonding after the lecture. The internal consistency and reliability of the survey results were measured using the Cronbach's alpha statistical analysis.[48] The value obtained (α > 0.8) corresponds to high acceptance reliability.

CONCLUSIONS

The COVID-19 pandemic has been a real challenge for teachers throughout the world. Social distance and remote classes required quick adaptation on the part of both students and teachers. Likewise, in this context huge areas of opportunity became apparent at the teaching level. In this work, a remote didactic approach was proposed, and successfully applied during the pandemic. This can now be adopted in contexts where the lack of appropriate equipment limits organometallic chemistry teaching. In this sense, computational calculations proved to be a useful and valuable tool for simulating a large number of spectra that, either because of technical limitations or due to the organometals' toxicity, would represent a challenge for undergraduate students. In this context, computational tools together with spectroscopic analysis constitute an easy, inexpensive, and efficient methodology for understanding complex phenomena at the undergraduate level. The advantages of computer-based methods enable the simulation of several IR spectra of different organometallic complexes within short time spans. We have demonstrated how IR spectra interpretation can easily be

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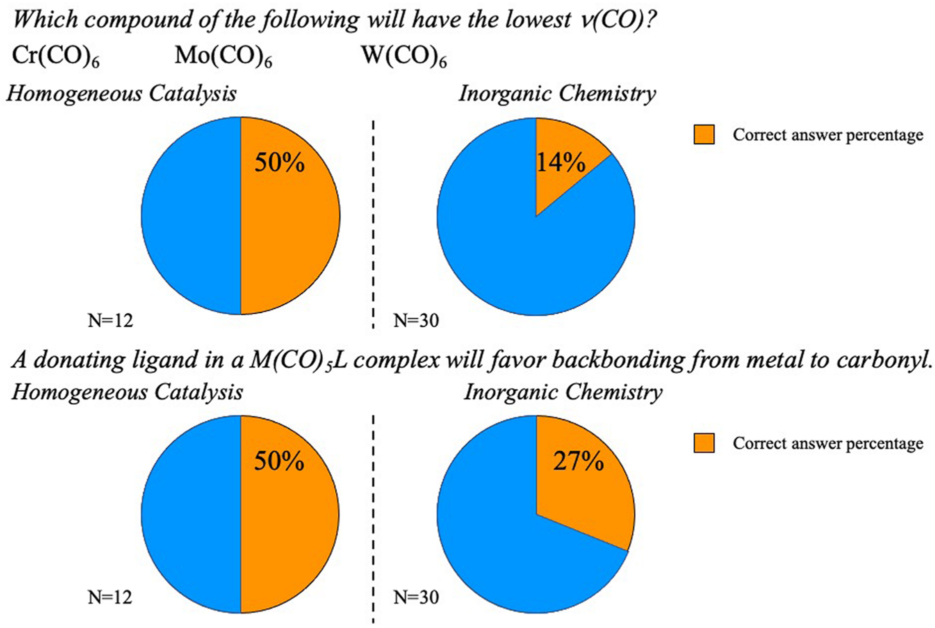

Figure 3. Diagnostic questions and students result.

Table 2. Evaluation after Lecture

Question	Correct Answers: Homogeneous Catalysis	Correct Answers: Inorganic Chemistry
Which compound of the following will have the highest ν(CO)? mer-$Mo(CO)_3(PPh_3)_3$; mer-$Mo(CO)_3(PCl_3)_3$; mer-$Mo(CO)_3(P(OMe)_3)_3$	58%	79%
Choose between the following complexes which one will have a lowest ν(CO). $[Co(CO)_4]^-$ or $Ni(CO)_4$	75%	83%
Choose between the following complexes which one will have a lowest ν(CO). $Mo(CO)_5(PH_3)$ or $Mo(CO)_5(PF_3)$	75%	85%
A donating ligand in a trans-$M(CO)_4L_2$ complex shift IR signal to higher ν(CO). True or false	58%	67%

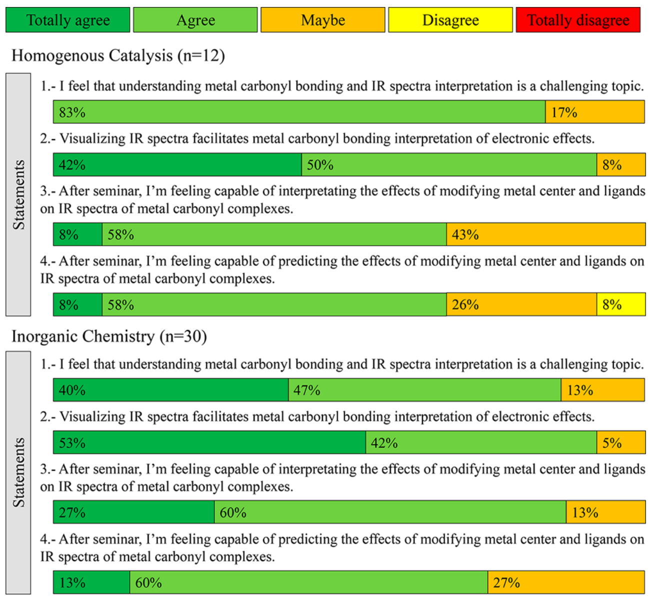

Figure 4. Student's exit survey analysis.

correlated with electronic effects between metal and ligands, clearly explaining how CO σ-donation and π-acceptance is modulated by the ligands that interact with the metal center. The proposed didactic approach, consisting of a lecture and a remote computational activity, proved to be a useful methodology for promoting chemical thinking. Two sets of students, second-year and upper-division students, were given a particular lecture format, where students filled the blanks, together with a remote computational activity enabling them to improve their ability to predict and interpret IR spectra. Even more importantly, they subsequently felt more confident to address this challenging topic, particularly as they were able to obtain and visualize the simulated spectra. This chemical thinking procedure, mobilizing different concepts, will surely aid students to understand metal ligand bonding. Through this remote pedagogical approach, students employed chemical thinking to solve a challenging task, by interconnecting different notions and tools. We firmly believe that the development of chemical thinking methodologies should become an area of increasing interest for teaching chemistry.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at https://pubs.acs.org/doi/10.1021/acs.jchemed.2c00577.

Evaluation and survey data containing the different questions and student's results on the evaluation and survey. (PDF, DOCX)

Academic content containing the filled lecture for the educators (PDF, DOCX)

Academic content blank containing the lecture with the blanks to fill by students and some important notes for the reader (PDF, DOCX)

IR videos containing the videos of the IR stretching (ZIP)

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■ AUTHOR INFORMATION

Corresponding Author

Antonio Reina – *Departamento de Química Inorgánica y Nuclear, Facultad de Química, Universidad Nacional Autónoma de México, 04510 Ciudad de México, México*; orcid.org/0000-0001-7527-7176; Email: antonio.reina.0711@quimica.unam.mx

Authors

Miguel Reina – *Departamento de Química Inorgánica y Nuclear, Facultad de Química, Universidad Nacional Autónoma de México, 04510 Ciudad de México, México*; orcid.org/0000-0003-4959-4105

Itzel Guerrero-Ríos – *Departamento de Química Inorgánica y Nuclear, Facultad de Química, Universidad Nacional Autónoma de México, 04510 Ciudad de México, México*; orcid.org/0000-0003-1741-9425

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jchemed.2c00577

Notes

The authors declare no competing financial interest.

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